

# Geodesia-KV: Graded and Query-Sparse KV-Cache Compression under Causal Evaluation

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## Abstract

KV-cache compression is usually evaluated along one of two axes: resident capacity for quantization and token eviction, or bytes fetched for sparse attention. Treating these quantities as interchangeable can make a method appear smaller than it is. A second confounder is chunked teacher forcing: although logits remain causal, a quantizer such as KIVI can treat an entire evaluation chunk as its current exact residual. We present a corrected, token-incremental study of Geodesia-KV, a family of training-free cache policies that combines monotone blockwise precision allocation, a small exact rate-distortion residual, and query-sparse reading using exact page summaries. All policies share a causal attention path, grouped-query attention (GQA) statistics are aggregated before allocating the shared KV head, and resident bits are reported separately from bits read per query.

At 16k context on WikiText-2, the approximately five-bit operating point improves on a local KIVI-4 port for Qwen2.5-3B (6.320 versus 6.322 perplexity at 4.96 versus 5.03 bits/value) and Qwen3-8B (10.540 versus 10.581 at 4.99 versus 5.03 bits/value). The margins are small. On a frozen six-book PG-19 transfer, the same 8B point loses (16.258 versus 16.147), so it is not a multi-corpus dominance result. Conversely, compressed-Quest is slightly worse than Quest on six WikiText-2 windows, but on six preregistered PG-19 books it improves perplexity (16.931 versus 17.022), resident rate (9.83 versus 16.25 bits/value), and read rate (1.95 versus 2.32 bits/value). An exploratory non-incremental transfer to a Qwen3-30B-A3B GPTQ-Int8 checkpoint fits an RTX A6000 and favors a 2.061-bits/value low-rate point, but is not primary evidence.

We additionally validate a packed, head-batched attention kernel in isolation. It reduces the represented cache by approximately  $6.9\times$  and is faster than a local dense `bmm` reference, but it is not yet integrated into Transformers’ `past_key_values`; therefore we make no end-to-end throughput or peak-VRAM claim. The resulting picture is mixed but actionable: the sparse branch transfers across corpora, the five-bit branch does not, and the remaining scientific and systems gaps are explicit.

## 1 Introduction

Autoregressive attention stores one key and one value for every layer and past token. Its memory therefore grows linearly with context length even when the model weights

have already been compressed. This creates two distinct resource problems. First, the *resident cache* must fit in device or host memory. Second, decode must *read* enough of that cache for every query. Quantization primarily attacks the former; query-sparse methods such as Quest [8] primarily attack the latter. Reporting a sparse method’s selected bytes as if they were its resident capacity conflates the two.

Evaluation can introduce a second conflation. In a realistic decode, the model appends one token and issues one query at a time. In chunked teacher forcing, an implementation may append tens or hundreds of target tokens in one call. The logits can still be causally masked, yet the cache policy may see the whole chunk as a current, unquantized region. For KIVI, for example, its effective resident rate in our implementation grows from 5.03 bits/value for a 64-token chunk to 5.69 for a 1024-token chunk. Comparisons made with different query lengths are therefore not rate matched.

This work began as an attempt to allocate a graded precision ladder to the entire context. The investigation uncovered and corrected future-query leakage, duplicated allocation of GQA-shared KV heads, a missing prefill observer, incorrect handling of partial Quest pages, contaminated teacher-forced retrieval, and resident/read accounting errors. These corrections materially changed the empirical conclusions. Rather than preserve an obsolete claim of universal dominance, we report the method that survives, the settings that do not transfer, and the missing system integration.

Geodesia-KV now denotes three related operating branches:

1. **Graded allocation:** every block remains represented on a monotone ladder of exact, scalar-quantized, or centroid states.
2. **Graded+RD:** a small exact residual is selected from the prompt by attention mass times squared value-reconstruction error.
3. **Compressed-Quest:** exact key boxes rank pages at query time, but the resident K/V pages themselves use the graded representation.

Our contributions are:

1. A causal,  $Q=1$  evaluation protocol with explicit separation of resident rate and read rate, model/text revisions, per-window results, and regression tests for query leakage and GQA sharing.

2. A monotone rate-distortion allocator over a heterogeneous precision ladder, including the exact scale overhead and a prefix constraint on demotion paths.
3. A target-free exact residual whose saliency combines prompt attention mass with value distortion, and a query-sparse variant that stores exact page boxes while compressing the underlying K/V.
4. Primary incremental results on 3B and 8B checkpoints, plus a frozen cross-corpus PG-19 test over the exhausted preregistered window list, with paired cluster-bootstrap intervals. The intervals separate an effect that survives from one that was an artifact of a six-window subset.
5. A numerically checked packed multi-head kernel microbenchmark and a measured account of resident context capacity, including the finding that grouped query attention makes device memory a non-binding constraint on maximum context for these checkpoints.
6. A record of negative quantum-inspired and conventional variants, including spectral K-V states, cumulant correction, Born-style gating, Hadamard rotation, VQ, static protection, and layer profiles.

## 2 Related Work

**Eviction and retained-token methods.** StreamingLLM [1] combines attention sinks with a recent window. H2O [2] retains heavy hitters. SnapKV [3] uses the final prompt queries to score tokens before generation. PyramidKV [4], PyramidInfer [5], CAKE [6], and Ada-KV [7] distribute eviction budgets across layers or heads. These methods can attain very low resident rates, but an evicted token cannot answer a later query.

**Query-sparse attention.** Quest [8] stores elementwise key minima and maxima for each page, bounds the largest possible query-key score, and reads top-ranked pages. InFLM [9], MInference [10], RetrievalAttention [11], and InfiniteHiP [12] use representative tokens, structured sparsity, approximate retrieval, or hierarchical pruning/offload. Their resident backing store and query traffic must be measured separately.

**KV quantization and merging.** KIVI [13] uses per-channel key quantization along time, per-token value quantization along channels, and an exact residual window. KVQuant [14] and GEAR [15] add non-uniform datatypes, sparse corrections, or low-rank structure. CaM [16], D2O [17], and MiniCache [18] merge tokens or exploit cross-layer similarity. Geodesia-KV combines scalar quantization and a centroid state under one rate constraint, then optionally adds an irregular exact residual.

**Architectural and systems alternatives.** DeepSeek-V2’s latent attention [19], Native Sparse Attention [20], Compressive Transformers [21], Infini-attention [22], and state-space models [23] change training or architecture. Our setting is training-free and targets frozen Qwen checkpoints [29, 30]. FlashAttention [24] supplies the online-softmax recurrence used by our prototype kernel. GPTQ [31] is used only for the weights of the exploratory 30B checkpoint; KV-cache rates remain independently measured.

**Long-context evaluation.** RULER [27] and LongBench [28] cover retrieval and downstream tasks that perplexity alone cannot establish. We use WikiText-2 and PG-19 for controlled language-model loss, and treat the absence of RULER/LongBench as a limitation rather than as evidence of memory fidelity.

## 3 Method

### 3.1 Causal policy state and GQA sharing

Let a decoder call contain queries  $q_0, \dots, q_{Q-1}$  and a cache prefix of length  $T$ . The representation serving  $q_0$  may depend only on the prefix and  $q_0$ ; using  $q_1, \dots$  to demote blocks leaks future teacher-forced information. Geodesia-KV therefore allows only the current query to trigger a demotion. Later queries may update statistics only for subsequent calls.

During prefill, a separate observer collects causal attention mass from the final prompt queries. This initializes both block masses and token saliency. For GQA, multiple query heads share one KV head. Their observed masses are aggregated first and the physical KV representation is allocated once. Allocating separately per query head both invents non-existent storage and can demote the same cache repeatedly.

### 3.2 Precision ladder and exact rate

The cache is partitioned into blocks of  $P = 64$  tokens. A block occupies one level

$$\mathcal{L} = \{16, 8, 4, 2, 1\},$$

where 16 is exact **bf16/fp16**, 8/4/2 are packed scalar quantizers, and 1 denotes a single centroid K/V pair repeated logically across the block. Keys are grouped along time and values along channels, following the asymmetry established by KIVI. Each level cost  $\beta_b(L)$  includes payload, offsets, scales, exact sink/window blocks, and any separate residual.

A newly closed block can only be demoted. It is never promoted from its reconstruction because the original values would no longer exist in a real compressed store. This monotonicity makes the simulated policy implementable without silently retaining a dense copy.

### 3.3 Equipotential allocation

Let  $m_b$  be the exponential-moving-average attention mass of block  $b$ , and let  $e_b(L)$  be a range/radius proxy for reconstruction error at level  $L$ . The allocator solves

$$\min_{\{L_b\}} \sum_b m_b^\gamma e_b(L_b) \quad \text{s.t.} \quad \sum_b \beta_b(L_b) \leq B, \quad (1)$$

where  $\gamma = 1$  in the final configurations. The Lagrangian relaxation decouples blocks through

$$m_b^\gamma e_b(L) + \lambda \beta_b(L).$$

Each possible one-step demotion has a marginal damage per saved bit. We sort these ratios and find the threshold that reaches the budget. A prefix rule is mandatory: reaching level 2 from 16 requires accepting the 16→8, 8→4, and 4→2 steps for that block. Without this constraint an early implementation reported budgets it had not actually paid.

### 3.4 Rate-distortion exact residual

Pure block allocation can miss a small number of high-impact tokens. For each prompt token  $j$ , the prefill observer records attention mass  $a_j$  and its distance from the value centroid of its block. The final residual score is

$$s_j = a_j \|v_j - \bar{v}_{b(j)}\|_2^2. \quad (2)$$

The highest-scoring prompt tokens are duplicated exactly. This selection is target-free at inference: it uses only prompt activations and no continuation labels. Its full 16-bit K and V cost is added to the graded base. The five-bit points use:

- Qwen2.5-3B: base  $B = 4.875$ , decay 0.25, exact fraction 0.78125%;
- Qwen3-8B: base  $B = 4.5$ , decay 0.5, exact fraction 3.125%.

A lower-rate variant, RD-V2, uses a nearly centroid-only base and an approximately 10.2% exact residual selected by the same value-distortion score. It is included only in non-incremental exploratory tables.

### 3.5 Compressed-Quest

Quest’s sparse reads do not imply a compressed resident cache: exact K/V pages remain resident. Our `box_sparse` branch stores, for every closed block, exact `fp16` key minima and maxima computed before quantization. For a query  $q$ , the box upper bound ranks all closed pages. The current partial page is always readable; no extra sink/window page is forced outside the top- $k$  set. Cold pages receive a hard mask.

The resident representation is the graded K/V cache plus all exact summaries:

$$R_{\text{resident}} = R_{\text{graded}} + \sum_b 2D \cdot 16 \text{ bits}. \quad (3)$$

The per-query read cost is all summaries plus the graded K/V of selected and current pages:

$$R_{\text{read}}(q) = R_{\text{boxes}} + \sum_{b \in \mathcal{S}(q) \cup \{\text{current}\}} R_b. \quad (4)$$

The final 8B setting uses base  $B = 10$ , decay 0.5, and a 12.5% page fraction. Exact boxes contribute approximately 0.25 bits/value but avoid ranking on quantized summaries.

### 3.6 Attention error certificate

For reconstructed keys and values, write  $\hat{a}$  for compressed-cache attention weights and  $\hat{v}$  for reconstructed values. The output error decomposes as

$$o - \hat{o} = \sum_j \hat{a}_j (v_j - \hat{v}_j) + \sum_j (a_j - \hat{a}_j) v_j.$$

With a score-error bound  $\varepsilon_b$  and value-error bound  $\delta_b^v$  for block  $b$ ,

$$\|o - \hat{o}\| \leq \sum_b \hat{a}_b \delta_b^v + \left( \frac{M^+}{M^-} - \frac{M^-}{M^+} \right) \max_j \|v_j\|, \quad (5)$$

where  $M^\pm = \sum_b \hat{a}_b e^{\pm \varepsilon_b}$ . The second term is capped by the universal  $\ell_1$  bound of 2. Validation runs with the dense reference enabled produced no observed violations. Hard sparse masking yields a valid but generally loose universal bound, so the certificate is currently more informative for graded reconstruction than for `box_sparse`.

### 3.7 Packed attention prototype

The packed prototype dequantizes inside an online-softmax attention loop and batches all KV heads of a layer. A separate reduction combines split-context partials. Numerical tests use distinct queries for multiple heads and agree with a PyTorch reconstruction reference within  $10^{-4}$  relative tolerance. This is a kernel prototype, not a Transformers cache implementation: quality experiments still retain dense K/V to simulate policies, and incremental append/demotion is not wired into `past_key_values`.

## 4 Experimental Protocol

### 4.1 Models, corpora, and hardware

Primary experiments use Qwen2.5-3B-Instruct revision `aa8e7253` and Qwen3-8B revision `b968826d`, both at a 16,384 token context on one NVIDIA RTX A6000. Activations are `bfloat16`. An exploratory scaling run uses community checkpoint `JunHowie/Qwen3-30B-A3B-GPTQ-Int8` revision `059b0db2` in `float16`; its weight quantization is separate from the KV policy. A Qwen3.5-0.8B checkpoint is retained only as

a diagnostic pilot because it predates the final incremental protocol.

WikiText-2 validation and test files are pinned by SHA256. For the final test file the hash is

```
bbf94c53a05abe9ee670d3b634360809
5822c85e26de37c70b24fc571964574a.
```

The 3B test uses six disjoint offsets from 0 to 81,920. The frozen 8B test uses six later offsets from 196,608 to 278,528.

PG-19 is fetched from `mozilla/pg19` at dataset revision `c021754c8e01c5b1cc83a1f549c1f97fbbb756b8`. Per-book boundaries and hashes are stored in manifests. The test text contains 12 books and has SHA256

```
42a0b97ff79ec96f53e18ec14bd770464
5bd10b7c50125935dbae721e9e8b445.
```

Test offsets are the preregistered per-book centers: each window is the 16,384 token span centered in one book. Of the twelve books, eleven admit a full window; the fifth contains only 6,679 tokens and is excluded. We report all eleven, so the PG-19 test list is exhausted and no window selection remains available. The offsets are 20,668, 90,369, 169,602, 246,823, 320,206, 402,659, 472,081, 529,495, 584,324, 659,151, and 736,259. An earlier version of this work reported the first six of these; Section 4.3 shows why that subset was insufficient.

## 4.2 Incremental perplexity

For each window we:

1. prefill the exact 16k prefix while collecting causal observations;
2. issue one query for each teacher-forced target token;
3. append that token to the same policy state;
4. accumulate the negative log-likelihood of the next token.

Every primary result uses  $Q = 1$  and 64 target tokens per window. Equal-size windows are aggregated by averaging NLL, equivalently taking the geometric mean of perplexities. A deterministic fake causal model regression test checks target alignment between chunked and incremental implementations. Real **bf16** full-cache PPL can differ by about 0.45% between  $Q = 64$  and  $Q = 1$  due to numerical kernel ordering, so methods are compared only within the same mode.

## 4.3 Paired confidence intervals

Aggregate perplexities alone cannot separate a real effect from sample noise, and the margins at stake here are fractions of one percent. Because every policy evaluated on a given window is teacher forced on exactly the same targets, the per-token negative log-likelihoods of two policies are

paired. Their mean difference is exactly the log ratio of the two perplexities,

$$\log \frac{\text{PPL}_a}{\text{PPL}_b} = \frac{1}{N} \sum_{i=1}^N (\ell_i^a - \ell_i^b), \quad (6)$$

so a confidence interval on the mean paired difference maps directly onto one for the perplexity ratio.

Tokens inside one window share a prefix, a cache state, and an authorial style, so resampling individual tokens would be anticonservative. We therefore use a cluster bootstrap that resamples whole windows with replacement (20,000 draws, percentile interval). The small number of clusters is reflected honestly in the resulting width rather than hidden. Runs are byte-reproducible: re-executing the eleven-window PG-19 test reproduces the previously published six-window aggregates of 16.147 and 16.258 exactly.

## 4.4 Metrics and baselines

We report:

- resident bits per scalar K or V value, including meta-data;
- bits read per value per query;
- teacher-forced perplexity;
- attention KL to the dense policy in separate diagnostic runs;
- represented cache bytes and attention-only latency for the prototype.

Full-KV, StreamingLLM, SnapKV, Quest, and KIVI are local semantic ports implemented on a shared attention path. They are not executions of the original version-pinned kernels. Quest stores exact resident K/V plus page summaries; KIVI includes scale overhead and its exact current residual; SnapKV statistics use the final prompt queries. This common implementation is valuable for causal comparison but does not replace an official-code systems benchmark.

# 5 Results

## 5.1 Primary five-bit comparison

Table 1 shows small local Pareto improvements. At 3B the PPL margin is 0.0026 (0.04%); at 8B it is 0.0408 (0.39%). These are not large-effect claims. The associated attention-fidelity runs are directionally consistent: at 3B KL is 0.00300 for Geodesia-KV and 0.00410 for KIVI-4; at 8B it is 0.00243 and 0.00390, respectively.

Table 1: Primary Q=1 WikiText-2 comparison, six windows and 64 targets per window. Rate includes payload, metadata, and exact residuals.

| Model | Method                | resident/read      | PPL            |
|-------|-----------------------|--------------------|----------------|
| 3B    | KIVI-4                | 5.031/5.031        | 6.3222         |
| 3B    | Geodesia-KV graded+RD | <b>4.961/4.961</b> | <b>6.3195</b>  |
| 8B    | KIVI-4                | 5.031/5.031        | 10.5807        |
| 8B    | Geodesia-KV graded+RD | <b>4.989/4.989</b> | <b>10.5399</b> |

## 5.2 Quest and compressed-Quest

On the full eleven-book PG-19 test compressed-Quest reduces resident rate by 39.5% and read rate by 15.4% relative to Quest while also improving perplexity. The paired ratio is 1.00516 with a 95% interval of [1.00076, 1.00945], which excludes one, and the sign is consistent in eight of eleven windows. This is the only comparison in this work whose three-axis advantage survives a frozen cross-corpus test with an interval attached.

The effect is nonetheless small: the lower bound is +0.08% of perplexity. What distinguishes it from the five-bit case is consistency across windows, not magnitude; both margins are below one percent. On WikiText-2 the sparse branch wins capacity, traffic, and diagnostic KL (0.04602 versus 0.04645) but loses PPL by 0.034 (0.34%), a cross-over that warns against selecting a sparse fraction on one corpus and claiming universal superiority.

## 5.3 Frozen cross-corpus test of the five-bit point

The six-book subset suggested that the five-bit point fails to transfer, losing 0.69% of PPL. Exhausting the preregistered list removes that conclusion (Table 3). Across all eleven books the margin shrinks to 0.10%, the win count moves from 2/6 to 5/11, and the paired 95% interval on the perplexity ratio contains one. On 693 paired targets the interval half-width is roughly 0.9%, an order of magnitude larger than the effect being measured.

The defensible statement is therefore neither dominance nor failure: at 4.989 against 5.031 bits/value the two policies have no measurable perplexity difference, so Geodesia-KV attains the same quality at lower resident capacity. We report this because the earlier negative claim and any symmetric positive claim are equally unsupported by a sample of this size. The same caution applies to Table 1, whose 0.04% and 0.39% margins have not yet been given intervals and are of the same order as the effect that dissolved here.

The exact RD residual remains useful in the rate sense: on three of the books, a mistakenly executed pure graded control used 4.486 bits/value and scored 24.307, whereas the corrected 3.125% residual used 4.985 bits/value and improved to 24.222. KIVI-4 scored 24.121 on the same books. The residual score in Equation 2 adds information,

but a globally fixed base/fraction has not been shown to beat KIVI-4 across corpora.

## 5.4 Exploratory low-rate and scaling results

The 30B-A3B checkpoint loaded and executed on a 48 GB RTX A6000, establishing feasibility of the requested model scale. RD-V2 is the best row in the six-window chunked screen. However, because these JSON files predate the `eval_mode` field and use Q=64, Table 4 must be regenerated incrementally before it supports a final scaling claim.

The 0.8B pilot is similarly diagnostic: pure graded points reach PPL 7.443 at 1.996 bits/value and 7.335 at 2.993 bits/value, while KIVI-2 reaches 8.905 at 3.033 and KIVI-4 7.570 at 5.034. Because the run used validation text, chunked evaluation, and old Quest accounting, it is excluded from primary tables.

## 5.5 Attention fidelity

Perplexity can improve when distant context is removed, acting as regularization rather than faithful memory. Table 5 partially separates these phenomena. It does not replace retrieval evaluation, but it prevents a lower PPL from being interpreted automatically as a more accurate cache.

## 5.6 Packed-kernel microbenchmark

The represented packed cache is materially smaller, and the fused attention-only kernel beats this particular dense reference (Table 6). The benchmark does not include model projections, MLPs, sampling, cache append, allocator execution, or Transformers dispatch. The quality runner still keeps dense K/V. Consequently the table is neither generation throughput nor measured end-to-end peak VRAM.

## 5.7 Resident context capacity

The preceding tables measure rate. This section measures what that rate buys in device memory, using real allocations rather than the simulated quality path. We load the model with 4-bit NF4 weights, measure the resident weight bytes and the peak activation of a genuine  $Q = 1$  decode step, and then locate the largest allocatable context by bisection. For Full-KV the search executes a real decode step at every probe, so its result is a demonstration rather than an estimate; for compressed formats we allocate buffers of the computed size, which establishes residency but not execution.

One number validates the accounting used throughout this paper. Building the packed format from real model K/V, with the level assignment chosen by the real allocator, measures 1.959 bits/value where the accounting reports

Table 2: Qwen3-8B, Q=1. “Wins” counts windows where compressed-Quest has lower PPL. WikiText-2 has six 64-target windows; PG-19 has the full eleven.

| Corpus     | Method                       | resident     | read/query   | PPL           | wins |
|------------|------------------------------|--------------|--------------|---------------|------|
| WikiText-2 | Quest                        | 16.250       | 2.317        | <b>10.238</b> | 4/6  |
| WikiText-2 | Geodesia-KV compressed-Quest | <b>9.854</b> | <b>1.957</b> | 10.272        | 2/6  |
| PG-19      | Quest                        | 16.250       | 2.317        | 16.043        | 3/11 |
| PG-19      | Geodesia-KV compressed-Quest | <b>9.836</b> | <b>1.961</b> | <b>15.961</b> | 8/11 |

Table 3: Frozen Qwen3-8B transfer to PG-19, Q=1, and the effect of exhausting the preregistered window list. The six-book column is the subset reported earlier; the eleven-book column is the full list.

|                            | six books     | eleven books       |
|----------------------------|---------------|--------------------|
| KIVI-4 PPL                 | <b>16.147</b> | 15.475             |
| Geodesia-KV graded+RD PPL  | 16.258        | <b>15.490</b>      |
| resident/read, KIVI-4      | 5.031         | 5.031              |
| resident/read, Geodesia-KV | <b>4.989</b>  | <b>4.989</b>       |
| Geodesia-KV wins           | 2/6           | 5/11               |
| PPL ratio                  | —             | 0.99904            |
| 95% CI                     | —             | [0.99046, 1.00795] |

1.990. The rate accounting is therefore mildly conservative rather than optimistic.

Table 7 supports a conclusion that is largely negative, and we state it rather than the flattering reading. Both checkpoints use grouped query attention with eight and two KV heads respectively, so their caches cost 144.0 and 36.0 KiB per token. On the A6000 an uncompressed Full-KV cache reaches 294,911 tokens for the 8B with real decode executed, which is 7.2× the model’s native window. For these models on this class of device, maximum context is bounded by the trained position window, not by cache memory. Compression changes the ranking without changing feasibility.

There is one practical exception. At the 131,072 token ceiling reachable with RoPE scaling, the 8B needs 23.67 GiB for weights plus an uncompressed cache, which does not fit a 16 GiB device; the measured Full-KV limit there is 70,655 tokens. The same window costs 7.99 GiB with the two-bit branch and 11.33 GiB with KIVI-4. Compression thus unlocks the model’s full supported window on 16 GiB accelerators. For the 3B no such point exists: its full window costs 6.42 GiB uncompressed, and even a 16 GiB device holds 413.7k Full-KV tokens, more than three times the reachable ceiling.

Comparing the two model families in Table 7 isolates what capacity depends on. Within a model the ratios between methods are fixed by rate alone, so Geodesia-KV 2-bit exceeds KIVI-4 by 2.44× at both 8B and 3B; what changes across models is only the absolute scale, through the 4× difference in bytes per token induced by two versus eight KV heads. Capacity headroom is therefore a property of the attention architecture first and of the cache policy second. A method that halves the

rate cannot compensate for a checkpoint that stores four times more per token, which is why the 3B reaches larger contexts than the 8B under every policy including the uncompressed one.

The axis on which compression is decisive is concurrency (Table 8), where the two-bit branch admits 8.8× more sequences than Full-KV and 2.5× more than KIVI-4 at the native window. Unlike the perplexity margins of Tables 1 and 3, these are multiplicative factors, not fractions of one percent. Note also that Quest is not competitive on this axis at all: keeping exact resident K/V plus page summaries costs slightly *more* than uncompressed storage, so its traffic advantage does not translate into capacity.

Two caveats bound these numbers. The KIVI and Quest per-token costs are projected from the rate accounting, since this work implements a resident packed format only for Geodesia-KV. And allocation feasibility is not usable context: beyond the trained window RoPE scaling would be required, which we do not evaluate here. Under a tight memory ceiling the projections are optimistic by 5–7% because ballast fragments the allocator, so they should be used with a margin.

## 6 Negative Results and Lessons

**Chunked evaluation as a false optimization signal.** Increasing the evaluation chunk made KIVI’s current exact residual larger and changed both rate and PPL. Final comparisons therefore moved to Q=1.

**Static prompt protection.** Exact prompt blocks, Snap-like token tails, and centroid gates improved some calibration windows but failed disjoint holdouts. A 2.061-bits/value exact-tail hybrid nearly matched SnapKV during development and then lost on holdout.

**Quantum-inspired spectral states.** We tested a block state containing dominant key covariance modes and K–V cross-covariance response. Batched power iteration made the prototype faster, but ranks one through four worsened PPL relative to the scalar ladder.

**Cumulant correction.** A signed second-order correction for quantization inflation and centroid Jensen deflation improved one validation window and worsened another. A single global correction does not transfer.

Table 4: Exploratory *chunked* Q=64 transfer of RD-V2. These rows are screening results, not primary Q=1 evidence. The 30B weights are GPTQ-Int8.

| Model   | Method            | resident     | read/query   | PPL           |
|---------|-------------------|--------------|--------------|---------------|
| 8B      | Full-KV           | 16.000       | 16.000       | 10.535        |
| 8B      | SnapKV            | 2.063        | 2.063        | 10.560        |
| 8B      | Quest             | 16.250       | 2.313        | <b>10.230</b> |
| 8B      | KIVI-4            | 5.043        | 5.043        | 10.539        |
| 8B      | Geodesia-KV RD-V2 | <b>2.061</b> | <b>2.061</b> | 10.517        |
| 30B-A3B | Full-KV           | 16.000       | 16.000       | 6.0008        |
| 30B-A3B | StreamingLLM      | 2.004        | 2.004        | 6.0341        |
| 30B-A3B | SnapKV            | 2.063        | 2.063        | 5.9900        |
| 30B-A3B | Quest             | 16.250       | 2.313        | 6.1513        |
| 30B-A3B | KIVI-2            | 3.051        | 3.051        | 6.2506        |
| 30B-A3B | KIVI-4            | 5.043        | 5.043        | 6.0600        |
| 30B-A3B | Geodesia-KV RD-V2 | <b>2.061</b> | <b>2.061</b> | <b>5.9413</b> |

Table 5: Diagnostic attention KL to dense attention. These runs isolate memory fidelity; they are distinct from PPL evaluation.

| Model | Method                       | resident      | KL             |
|-------|------------------------------|---------------|----------------|
| 3B    | KIVI-4                       | 5.086         | 0.00410        |
| 3B    | Geodesia-KV graded+RD        | <b>4.990</b>  | <b>0.00300</b> |
| 8B    | KIVI-4                       | 5.086         | 0.00390        |
| 8B    | Geodesia-KV graded+RD        | <b>4.989</b>  | <b>0.00243</b> |
| 8B    | Quest                        | 16.250        | 0.04645        |
| 8B    | Geodesia-KV compressed-Quest | <b>10.238</b> | <b>0.04602</b> |

**Born-style projection.** Selecting the smallest block subspace containing a target fraction of centroid softmax mass improved several validation windows, then failed the frozen test. Composing it with RD-V2 was slower and worse than RD-V2 alone.

**Hadamard rotation and key/value asymmetry.** Hadamard incoherence processing [25, 26] spread key outliers into channels already protected by per-channel scales and degraded quality. Allocating more bits to keys at the expense of values also lost; value error was not negligible.

**Vector quantization and fine groups.** Multiple centroids help only below approximately 1.5 bits/value. Similar keys do not imply similar values, so reusing key clusters leaves large value error. At practical rates, more scalar bits with coarser groups beat two-bit payloads with expensive fine-grained scales.

**Uniform layer profiles.** Several six-group layer profiles were tested at matched mean capacity. Uniform allocation was better or indistinguishable. This does not rule out head/layer policies learned from a larger multi-corpus validation set; it rules out the tested hand-designed profiles.

## 7 Limitations

**Small evaluation sample.** The PG-19 test now uses all eleven preregistered windows and reports paired intervals, but 693 targets still give a 95% half-width near 0.9% of the perplexity ratio, while the effects at stake are 0.1–0.7%. The WikiText-2 tables remain at six windows and have no intervals at all; their margins should be read as unresolved rather than established. Because the PG-19 list is exhausted, added power must come from further corpora, not from further windows of this test.

**Local baseline ports.** Shared semantics expose causal bugs and allow matched accounting, but original baseline releases and optimized kernels must still be executed where possible.

**Incomplete task coverage.** No final RULER, Long-Bench, or multi-key retrieval suite is reported. Perplexity and attention KL are insufficient to establish long-term retrieval fidelity.

**No real compressed model cache.** Quality runs simulate compressed states while retaining dense K/V. The packed kernel is isolated and supports neither real incremental append nor integration with `past_key_values`. We therefore do not claim end-to-end speed or serving throughput. The capacity results of Section 5.7 measure real allocation and, for Full-KV, real decode, but the concurrency counts are cache-only arithmetic rather than a served workload.

**Mixed cross-corpus result.** Compressed-Quest transfers on PG-19 with an interval that excludes one, but misses WikiText-2 PPL. The five-bit point is statistically undecided against KIVI-4 on PG-19. No current point beats every baseline on every measured axis from 3B upward, and on the perplexity axis several comparisons are simply below the resolution of the available sample.

**Exploratory 30B protocol.** The 30B checkpoint fits and produces favorable chunked results, but Q=1 incremental replication, official baselines, and system measurements are still required.

Table 6: Head-batched attention-only microbenchmark at 16k. Dense is a local `bmm` reference, not FlashAttention or PagedAttention. Times include all KV heads of one layer and are multiplied by the number of serial layers.

| Model      | packed GiB | dense GiB | compression | packed ms/token | dense ms/token | speedup |
|------------|------------|-----------|-------------|-----------------|----------------|---------|
| Qwen2.5-3B | 0.0815     | 0.5625    | 6.90×       | 3.17            | 11.90          | 3.75×   |
| Qwen3-8B   | 0.3257     | 2.2500    | 6.91×       | 9.93            | 12.44          | 1.25×   |

Table 7: Maximum single-sequence context with NF4 weights resident. The 16 GiB row is measured by ballasting the device down to that ceiling and repeating the real bisection; other rows are projected from the measured weight and activation cost. Native windows are 40,960 (8B) and 32,768 (3B); both extend to 131,072 with RoPE scaling.

| VRAM   | Qwen3-8B, 144.0 KiB/token |        |                   | Qwen2.5-3B, 36.0 KiB/token |        |                   |
|--------|---------------------------|--------|-------------------|----------------------------|--------|-------------------|
|        | Full-KV                   | KIVI-4 | Geodesia-KV 2-bit | Full-KV                    | KIVI-4 | Geodesia-KV 2-bit |
| 16 GiB | 76.0k                     | 241.8k | 590.3k            | 413.7k                     | 1.32M  | 3.21M             |
| 24 GiB | 134.8k                    | 428.7k | 1.05M             | 648.7k                     | 2.06M  | 5.04M             |
| 48 GiB | 311.1k                    | 989.2k | 2.41M             | 1.35M                      | 4.30M  | 10.51M            |
| 64 GiB | 428.6k                    | 1.36M  | 3.33M             | 1.82M                      | 5.80M  | 14.16M            |

Table 8: Concurrent Qwen3-8B sequences whose caches fit alongside NF4 weights on a 48 GiB device. KV cache only; a real serving engine adds per-sequence activation and scheduler state.

| Method            | at 40,960 | at 131,072 |
|-------------------|-----------|------------|
| Full-KV / Quest   | 6         | 2          |
| Compressed-Quest  | 11        | 3          |
| KIVI-4            | 21        | 6          |
| Geodesia-KV 2-bit | <b>53</b> | <b>16</b>  |

## 8 Conclusion

Geodesia-KV is best understood as a measured family of rate-distortion and query-sparse cache policies, not as a completed universal replacement for every KV technique. The causal Q=1 protocol establishes small WikiText-2 improvements over a local KIVI-4 port at 3B and 8B, which currently lack confidence intervals. On the exhausted PG-19 window list the five-bit point is statistically undecided against KIVI-4, delivering equal perplexity at lower resident capacity; the earlier claim that it fails to transfer was an artifact of a six-window subset. In the complementary sparse regime, compressed-Quest reduces resident storage and query traffic and improves PG-19 perplexity with an interval that excludes one, though it does not improve WikiText-2 perplexity.

The capacity measurements reframe what these rates are worth. Because grouped query attention has already made the cache small, device memory does not bound maximum context for either checkpoint, except when a 16 GiB accelerator must hold the 8B at its RoPE-extended window. Compression instead multiplies concurrency, by 8.8× over Full-KV at the native window. That is the axis on which the measured advantage is a factor rather than

a fraction of a percent, and it is where we would direct further work.

The next algorithmic step is not further tuning on these test windows. It is a layer/head allocation policy calibrated only on multi-corpus validation and then frozen for a larger paired test, sized in advance for the effect it needs to resolve. The next systems step is a true packed `past_key_values` implementation with append, partial blocks, GQA sharing, monotone demotion, and sparse page reads. Only after comparison with FlashAttention/PagedAttention and official baseline kernels should end-to-end speed and throughput claims be made.

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